



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI INHA UNIVERSITY IN TASHKENT

Module Name: Creative Engineering Design
Module Code: SOC1020
Project Type: Group Project (4-6 students)
Project Assigned Date: 17.04.2023
Team Registration Link: [Click Here](#)
Team Registration Deadline: 01.05.2023 (midnight)
Deadline: 07.05.2023 (midnight)
Weighting: 20%

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Office Hours: Mon, 13:00 - 14:00
Fri, 13:00 - 14:00
Office: Room B411

Learning Outcomes

1. Understanding key steps in product design and development process
2. Understanding the basic human factors involved in the design of physical products
3. Understanding the importance of universal design and accessibility in product development
4. 3D modelling of a conceptual design of a product and refining it for further development
5. Applying TRIZ principles in finding solutions to contradicting technical problems
6. Preparing technical documentation for proposed conceptual design

Things to Submit:

1. **PRINTED and BOUND PRESENTATION SLIDES** as specified in this assignment
2. **PRINTED and SIGNED MEETING MINUTES** for each meeting conducted along with the supplementary materials used or created during the meeting (e.g. customer interview transcripts, surveys, handwritten sketches, literature review, competitor products, TRIZ resolutions, technical drawings and documentation).
3. **ARCHIVE FILE** containing the electronic version of your presentation slides and all supplementary materials (e.g. customer interview transcripts, surveys, handwritten sketches, literature review, competitor products, TRIZ resolutions, technical drawings and documentation, Autodesk Fusion 360 files (.f3d files), or any other relevant findings or 3D models used) **submitted to eClass.**

Design Project Overview

You are required to create the conceptual design of a discrete physical product (e.g. consumer electronics, home appliances, personal care, stationary, garden tools, kitchenware, car accessories, mobilephone accessories, products for pets, etc) that illustrates key aspects of user interaction and working principle of your proposed product. Central to this project is a team-based approach to conceive and design a new physical product and present its 3D model in the presentation session. The goal of this project is to learn integral principles of the design of everyday things, to improve teamwork skills and to appreciate the inherent multidisciplinary nature of product development. You can pick an idea for your physical product from the list provided in Scenario section below. If your team has a different idea for a physical product which is not listed below, please validate your proposal with me by email or contact me during my office hours.

Scenario:

You have been employed by the manufacturing company X to develop the concept of a physical product they are going to mass produce and launch to the market next year. You are asked to collect customer needs, research competitor products, compile product specifications, generate multiple concepts, select the most promising concept for the development of a detailed product specification in terms of its parts and assembly in a 3D sketch. Your key findings on completed user research and the final version of a 3D sketch should be presented to the top management of the company X as the final output of your project.

You can select the area of business of the company X from given list below, or come up with your own idea. Your product should be innovative and must be useful in treating one of the problems listed below:

- Improving the quality of life for people with disabilities using the key principles of universal design (e.g. patients suffering dementia, paraplegic, bad eyesight or hearing, cognitive impairment, or psychological problems etc) or elderly.
- Heating or cooling houses in a cheap and energy efficient way.
- Cooking, storing, transporting or serving food in a cheap and eco-friendly way.
- Maintaining personal hygiene in a cheap and eco-friendly way.
- Simplifying the use of hand tools in constructions and maintenance.
- Helping athletes in reaching their goals in sports.
- Helping musicians in rehearsing or performing on stage.
- Traveling outdoors or supporting active lifestyle for office workers.
- Doing the garden work, watering flowers, cultivating plants.
- Making remote control of cars, TV, air conditioner or gaming consoles more user-friendly. Find creative and intuitive ways of mapping buttons to actions, and providing comprehensive feedback on the result of chosen action via screen or other means.

There should be a demonstrable market for the product. One good way to verify a market need is to identify existing products that attempt to meet the need. Your product need not be a variant of an existing product, but the market need addressed by your product should be clearly evident. The product does not need to have a tremendous economic potential, but should at least be an attractive opportunity for an established firm with related products and/or skills.

You should be confident of being able to prototype the product for a short period of time. Therefore you are urged to concentrate your efforts in specific needs of a small target market. Avoid addressing too many needs of big and complex markets. Moreover, you are not encouraged to choose product ideas which would assume a technological breakthrough (e.g. nanobots which can cure cancer) or unattainable market (e.g. people who live in Greenland and own penguins). Try to be realistic on both your market and technology for your proposed project. You should have access to

more than five potential users of the product (more than 20 would be nice). You may also establish ties with an existing company that may be interested in your concept at the end of your project.

How to Register Your Team?

If you want to change your team for Project 2, you must update your team in below link before the deadline:

Team Registration Link: [Click Here](#)

Teams cannot change their members after the deadline (1 week before official deadline) has passed. It is team leader's responsibility to ensure that their team members up-to-date in above spreadsheet.

Teams must contain at least 4 students and at most 6 students. Students from different sections cannot join the same team at the same time. This is due to the differences in their timetables which can be an issue while their project presentation.

What if I cannot find a team?

I will automatically assign those who could not find teams into new teams created by me. It is the responsibility of each student to check Google Spreadsheet to learn about his/her team after this assignment. Students who fail to participate in this team's activities will **get 0** for this assignment.

How to Complete Your Project?

You should organize group meetings every 2-3 days to assign tasks to members of the team and crosscheck on the progress of your project. Your meetings should be documented by filling out a **meeting minutes document** enclosed to this assignment. Below are the important steps in your project:

1. Register/update your team.
2. Formulate your problem statement, assumptions and expected outcomes.
3. Plan your project's major tasks using Gantt chart
4. Research customers and competitor products. Identify, interpret and rank key customer needs by their relative importance.
5. Define a list of design metrics that your product can be evaluated with in a most objective way.
6. Investigate competitor products in the market and find out their technical characteristics from the perspective of your design metrics and compare them to one another.
7. Identify key design metrics that will make the most impact to customers using QFD matrix.
8. Compile the final list of your product's target specifications in terms of ideal and marginal values for your design metrics resulted from benchmarking competitor products and completing QFD analysis.
9. Generate your concept sketches on paper using techniques such as storyboarding, SCAMPER, analogy, worst possible idea, brainstorming, mind mapping, morphology. Ensure that these concepts meet your product's target specifications. Apply decision matrix to select most promising ones and iterate at least 2-3 times.
10. Research literature on technical feasibility of your product, identify trade-offs among your design metrics (i.e. the roof of QFD house) and resolve these trade-offs using TRIZ guidelines.
11. Develop high-fidelity 3D model of your final product using a CAD software. Explain its working principle through the assembly of its parts in a CAD software. Your product should consist of at least 2-3 parts.

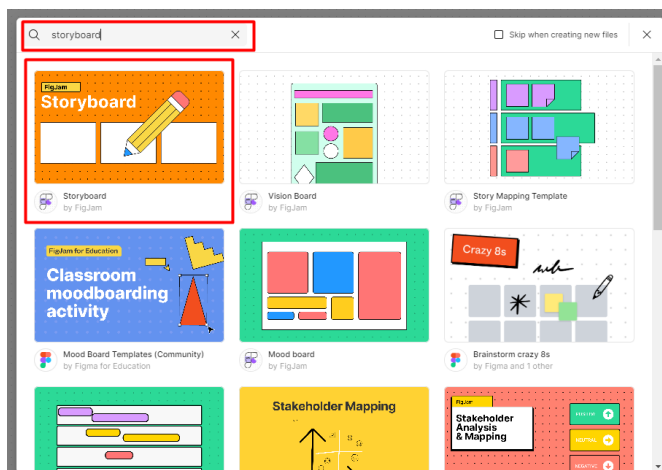
12. Reflect on your design process. Find strong and weak points of your work flow and proposed concept.
13. Prepare presentation slides and technical drawings.
14. Submit electronic version of your presentation file, meeting minutes and all supplementary materials to eClass before the deadline.
15. Print and bind your presentation file and meeting minutes along with relevant supplementary materials.

How to Document Meetings?

You can conduct meeting in two formats: online or offline. Each meeting has to be documented in an appropriate way as instructed below.

Online Meetings

Online meetings can be conducted in FigJam where you can brainstorm different parts of your project. FigJam has wide range of templates for creating Events Calendar, Personas, Storyboards, User Journey Map, Problem Statement, Prioritization Matrix, Product Specifications, Competitor Analysis, Flow Charts and UI Wireframes. In order to find the right template, click  button, and search for a corresponding template:



Meeting Agenda	
TODAY'S DATE	
Meeting Summary This is a paragraph of things to give a summary of the meeting planned	
Action Items: 1. This is item 1 2. This is item 2 3. This is item 3	
Notes & Next Steps: This is a paragraph of notes and next steps beyond the meeting.	

Before each meeting, team leader is responsible to place above clipboard to the FigJam canvas and fill in the form. This meeting will be considered as an appropriately documented meeting.

Offline Meetings

Offline meetings should be documented by filling out a form attached in Appendix 1. Team leader has to prepare this meeting minute document before the meeting and ask each participant to sign to prove their presence. Also team leader should print out necessary templates provided in Appendices 2 – 6 for the upcoming meeting, so team members can discuss and fill out these templates in the process. All work produced during this offline meeting has to be stapled to meeting minutes document (as a cover page of this meeting) and added to your supplementary materials. You should submit them along with your presentation slides during your project's presentation.

How to Submit Your Project?

You should archive your presentation slides and supplementary materials into a single archive file named as your team name (e.g. Lions.zip). This file should be submitted **only ONCE** by the team leader to eClass **before deadline**.

You should print out your presentation slides and bind them to a folder. For each meeting you have conducted with your team members, you should also print a meeting minutes document signed by attendees. Anything relevant used or created during your meeting should be enclosed to your meeting minutes document. All your printed and signed meeting minutes and their relevant attachments must be bound to a folder and submitted as supplementary materials to your presentation slides. **All your printed documents should be given to me first class after deadline.**

How to Prepare Presentation Slides?

You will not present your project. You should prepare presentation slides only.

The structure of your presentation should be as shown below:

1. **Title page** contains team name, project title, team members' names and student IDs, section number.
2. **Mission statement page** contains the formulation of your goal, assumptions and expected benefits/objectives.
3. **Introduction page(s)** provides context for your product. This may involve brief explanation of historical evolution of the product or technology, typical use cases of the product.
4. **Customer needs page(s)** contains key highlights of your user research process and the list of prioritized customer needs.
5. **Product specifications page(s)** contains design metrics for your product and their corresponding values for competitor products. It also has to show your QFD analysis and key design metrics where you should beat your competitors. Final list of target values you're your product's design metrics should be provided in a separate page.
6. **Concept generation page(s)** contains key highlights of your concept generation process and featured concept sketches you have initially proposed. You should highlight the use of concept generation techniques like SCAMPER, morphology, mind mapping, etc.
7. **Concept selection page(s)** contains methods of evaluating proposed concepts through decision matrices and iterations.
8. **TRIZ resolution page(s)** contains key technical trade-offs among your design metrics and the use of TRIZ guidelines in their resolution.
9. **Technical drawings** demonstrate the working principle of your proposed product through high-fidelity 3D model designed using CAD software. It should illustrate individual parts of your proposed product and the way they should be assembled or operated.
10. **Universal Design** demonstrates the compliance of proposed design with Universal Design principles and human factors. It should provide use cases on how proposed concept can be operated by people with different abilities.
11. **Conclusion page** discusses about challenges and weak parts of the proposed concept. It also provides reflection on the entire process of your team's work as a product development team.
12. **References page** provides links to relevant literature, 3D models, and all other online resources used in designing your product's sketch.

Online Learning Resources

Fusion 360 video tutorials

- [Official Fusion360 Courses](#)
- [Fusion 360 in 30 minutes](#)
- [Fusion 360 tutorials in Youtube \(Russian\)](#)

3D Models Library

- www.thingiverse.com
- www.youmagine.com
- www.myminifactory.com

Project Evaluation

In general, projects will be evaluated along the following aspects that contribute to the overall quality of the accomplished work:

- Compliance with the product design process
- Feasibility of proposed product with current technology and the market
- Reliability and thoroughness of customer needs research
- Adequacy of design specifications and QFD
- Evolution of concepts from quantity to quality
- Originality of the proposed product
- Quality of the presentation
- Quality of the supplementary materials

The marks for this project will be distributed according to following scheme:

Assessment Component	Mark (points)
Introduction and mission statement	5
Customer needs	10
Design variables	10
QFD and Competitor Benchmarking	10
Target product specifications	5
Concept generation and selection process	10
TRIZ resolutions	10
Ergonomics and universal design	10
Technical Drawings and Documentation	10
Quality of presentation slides	10
Quality of supplementary materials	10
Total	100

Penalties

Teams are subject to a penalty or disciplinary action for reasons of academic dishonesty and negligence. Academic dishonesty can be any attempt by a student to submit 1) work completed by another person outside the team without proper citation, 2) or give improper aid to another student outside the team in the completion of an assignment. Academic negligence can be disregarding the key requirements of the project including the timely submission of the project's deliverables. Following penalties will be exercised if any of the below cases happen:

- **Late submission.** If you missed the deadline, you must send your project deliverables over email to s.abdullaev@inha.uz within the next day after the deadline. You will be penalized by **20 points** for late submission. **Important notice:** you are not required to send your project deliverables by email if you have already submitted them to eClass before the deadline. If you submit your project deliverables to 2 places: eClass and email, you will get a penalty of **5 points** for confusing the assessment.
- **Not informing about team change.** If you change a team or unable to find a team, you should inform me by email about this fact before team registration deadline. I will make changes in my teams list or assign you to a different team. If you fail to inform this fact or inform it after the deadline, you will be penalized by **10 points**.

Any other academic misconducts including plagiarism, cheating, or breaching internal rules and regulations of IUT will be brought to the attention of Academic Committee for Penalty and Awards.

Disclaimer:

- I reserve the right to grade each member of the team separately if I have a suspicion on his/her contribution to the project. I will review meeting minutes submitted with the project, and if I find a member who has not contributed to the project, I will call this member for a short viva regarding his/her contribution. If member fails to show up, he/she will receive 0 for the project.
- I reserve the right to call the whole team for a short viva if I have suspicion about the quality of work submitted. If team fails to explain how web application has been implemented, each member of the team receive 0 for the project.